25901195 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECCSC200 - FOUNDATIONS OF DATA SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define secondary data [K] / [CO1]
2. Find the mode of: 4, 6, 6, 9, 11. [K] / [CO2]
3. Identify whether the events “getting head” and “getting tail” are mutually exclusive. [K] / [CO3]
4. Describe the different types of averages used in statistics. [U] / [CO2]
5. Write the formula for conditional probability. [U] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Distinguish between diagrammatic presentation and graphic presentation of data. [K] / [CO1]

7. Write a brief note on central tendency and the purpose of average. Find the arithmetic mean of: 14, 18, 22, 26. [U] / [CO2]

8. Define a random experiment. Explain the term clearly and give one suitable example. [K] / [CO3]

9. Describe the **direct personal investigation method** of collecting primary data [K] / [CO1]

10. Briefly explain Arithmetic Mean and find the average of the marks obtained by 6 students: 40, 45, 50, 55, 60, 65. [K] / [CO2]

11. State the addition theorem of probability for three events. Write the formula and explain the meaning of each term. [K] / [CO3]

12. State the axioms of probability. Explain each axiom briefly. [K] / [CO3]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Distinguish between diagrammatic presentation and tabular presentation of data. [U] / [CO1]

14. Define Mode. Explain its merits and demerits and compare it with Mean and Median. [K] / [CO2]

[15x1 = 15]